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**PROJECT
REPORT**

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Online Food Delivery Business Intelligence & Performance Analysis Using SQL

Introduction

This project focuses on analyzing an online food delivery platform using SQL to extract meaningful business insights from structured relational data. The dataset represents the core operational components of a food delivery business, including customers, restaurants, orders, order items, and delivery agents. The objective is to evaluate business performance, understand customer behavior, and assess operational efficiency.

The analysis is important because online food delivery businesses operate in a highly competitive environment where data-driven decisions are critical. Understanding revenue trends, customer retention, restaurant performance, and delivery efficiency helps improve profitability and service quality. SQL was used extensively to join tables, aggregate metrics, and derive insights relevant to real-world business scenarios.

Problem Statement

Despite having large volumes of transactional data, the business lacks clear visibility into customer behavior, revenue performance, restaurant growth, and delivery efficiency. The data is spread across multiple tables, making it difficult to generate actionable insights.

This project aims to answer key business questions related to revenue drivers, customer retention, restaurant contribution, and operational bottlenecks. Solving these problems enables better decision-making, improved customer experience, and sustainable business growth.

Business Objectives

- **Evaluate Revenue Performance:** Analyze total revenue, average order value, and revenue trends over time to identify high-performing periods, cities, and restaurants contributing most to overall business growth.
- **Improve Customer Retention and Engagement:** Measure customer ordering frequency, repeat purchase behavior, and signup-to-order patterns to identify loyal customers and detect churn risks.
- **Assess Restaurant Performance and Growth Opportunities:** Identify top-performing and underperforming restaurants based on order volume, revenue contribution, ratings, and location to support restaurant partnership and expansion strategies.
- **Analyze Product and Order Item Performance:** Determine the most frequently ordered items, high-revenue food items, and order composition patterns to guide menu optimization and pricing decisions.
- **Optimize Discount and Pricing Strategies:** Evaluate the impact of discounts on order value and revenue to ensure promotional strategies increase sales without negatively affecting profitability.
- **Enhance Delivery and Operational Efficiency:** Measure delivery times, delivery agent performance, and city-wise delivery efficiency to identify operational bottlenecks and improve service quality.

- **Understand Payment Method Preferences:** Analyze payment method usage across orders to support payment optimization, reduce friction, and improve customer checkout experience.
- **Enable Data-Driven Decision Making Using SQL:** Transform raw transactional data into actionable insights through SQL-based analysis, supporting strategic planning and operational improvements across the business.

Dataset Overview

The dataset represents a structured relational database for an online food delivery platform, designed to capture end-to-end business operations. It includes customer information, restaurant details, order transactions, item-level order data, and delivery agent performance. The dataset enables comprehensive analysis of revenue, customer behavior, restaurant growth, and operational efficiency using SQL.

Tables Involved

The dataset consists of the following core tables:

- **Customers** – Stores customer demographics, location, and signup information.
- **Restaurants** – Contains restaurant details such as city, cuisine type, and ratings.
- **Orders** – Central transaction table linking customers and restaurants, including order value, discounts, payment method, and delivery time.
- **Order_Item** – Item-level breakdown of each order, including food items, quantities, and prices.
- **Delivery Agents** – Stores information about delivery personnel, their city, joining date, and performance ratings.

Record Size

- The dataset contains multiple records per table, with one-to-many relationships between customers and orders, orders and order items, and restaurants and orders.
- Exact row counts are not fixed, allowing the dataset to scale and simulate real-world business volume for analytical practice.

Time Period

- The data spans multiple dates, including customer signup dates, order dates, and delivery agent joining dates.
- This allows for time-based analysis such as trends, growth patterns, and performance comparisons over different periods.

Type of Data Stored

- **Transactional Data:** Orders, order items, payments, discounts, and delivery times.
- **Master Data:** Customers, restaurants, and delivery agents.
- **Performance Metrics:** Ratings, revenue values, order amounts, and delivery efficiency indicators.

ER Diagram Explanation

The ER diagram represents the core entities and their relationships within the online food delivery database, showing how business operations are connected through structured data.

Entities

The main entities in the database are Customers, Restaurants, Orders, Order_Item, and Delivery Agents. Customers store user details, restaurants store food provider information, orders capture transaction-level data, order items store individual food items per order, and delivery agents represent the delivery workforce.

Relationships

- A Customer places Orders.
- A Restaurant receives Orders.
- An Order contains multiple Order_Items.
- Delivery Agents are associated with orders indirectly through delivery performance metrics.

Cardinality

- One Customer can place many Orders, but each order belongs to one customer (1:N).
- One Restaurant can receive many Orders, but each order is linked to one restaurant (1:N).
- One Order can contain many Order_Items, but each order item belongs to one order (1:N).

Key Constraints

- Primary keys uniquely identify each entity (e.g., customer_id, restaurant_id, order_id, agent_id).
- Foreign keys enforce relationships, such as customer_id and restaurant_id in the Orders table and order_id in the Order_Item table.
- These constraints ensure data integrity and prevent orphan records.

Overall, the ER diagram visually demonstrates how customers, restaurants, orders, items, and delivery operations are interconnected to support analytical queries and business insights.

Key SQL Analysis

```
-- 1. Total revenue generated by the food delivery app
SELECT SUM(order_amount - discount) AS total_revenue FROM orders$;
```

1. Business Question

What is the actual revenue generated by the company after accounting for discounts applied on customer orders?

2. Tables Used

- **Orders:** This table contains transactional data including order amount and discount applied per order.

3. Business Insight Derived

This query calculates the net revenue, not just gross sales. By subtracting discounts from order amounts before aggregation, it provides a realistic view of how much revenue the business actually earns. This metric is critical for evaluating financial performance, pricing strategies, and the effectiveness of discount campaigns.

Data Analysis & Insights

Based on the SQL analysis and observed query outputs:

Major Trends

- Revenue shows consistent contribution from repeat customers, indicating stable demand.
- Orders are concentrated in specific cities, highlighting geographic revenue hubs.
- Restaurants with higher ratings tend to receive more frequent orders.

Patterns

- Discounted orders increase order volume but reduce per-order revenue.
- Digital payment methods dominate, indicating customer preference for cashless transactions.
- Certain cuisines consistently outperform others across cities.

Customer Behavior

- Customers who place repeat orders contribute a significantly higher share of total revenue.
- Early signup customers show higher long-term engagement.
- Order frequency increases during specific time periods, indicating peak demand windows.

Revenue Distribution

- A small percentage of restaurants generate a large portion of total revenue.
- High-value orders often include multiple items rather than single-item purchases.
- Revenue is unevenly distributed across cities, indicating expansion opportunities.

Performance Insights

- Faster delivery times correlate with higher customer ratings.
- Delivery agents with longer tenure tend to maintain better ratings.
- Operational delays negatively affect repeat customer behavior.

Performance Optimization

Indexes were applied on primary and foreign keys to improve query speed. Efficient joins, CTEs, and optimized aggregations ensured scalability and performance for large datasets.

Key Findings Summary

1. Discounts significantly impact net revenue
2. Repeat customers drive revenue
3. Few restaurants dominate sales
4. Faster delivery increases satisfaction
5. Ratings correlate with order volume
6. Certain cities act as revenue hubs
7. Multi-item orders increase value
8. Experienced agents perform better

Business Recommendations

Revenue Growth

- Optimize discount strategies by targeting high-value and repeat customers.
- Promote top-performing restaurants and high-demand cuisines.
- Encourage bundle and combo orders to increase order value.

Customer Retention

- Introduce loyalty programs for repeat customers.
- Improve service quality in cities with lower repeat rates.
- Personalize offers based on past ordering behavior.

Cost Efficiency

- Optimize delivery routes in high-order-density cities.
- Reduce excessive discount usage that does not improve retention.
- Focus incentives on high-performing delivery agents.

Operational Improvement

- Monitor delivery time KPIs closely.
- Provide training and incentives for underperforming agents.
- Use performance data to improve restaurant onboarding standards.

Conclusion

This project demonstrates how SQL can transform raw data into actionable business intelligence. It highlights strong analytical skills, real-world problem-solving, and data-driven decision-making.

Future Scope

Future enhancements include BI dashboard integration, automated reporting, real-time analytics, advanced forecasting models, and machine learning-based predictions.